

Notes: Hansen (Econometrica, 1982)

Large Sample Properties of Generalized Method of Moments Estimators

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1 Big picture

- **Question.** How can we estimate and test econometric models when the model is most naturally expressed as population orthogonality conditions rather than as a fully specified likelihood?
- **Core object.** The model implies moment restrictions

$$E[f(x_t, \beta_0)] = 0,$$

where x_t is observed, β_0 is the parameter vector, and f collects the orthogonality restrictions.

- **Estimator.** The generalized method of moments estimator chooses β so that the sample moments

$$g_N(\beta) = \frac{1}{N} \sum_{t=1}^N f(x_t, \beta)$$

are as close to zero as possible under a chosen quadratic metric.

- **Main consistency result.** Under stationarity, ergodicity, continuity, identification, and appropriate weighting-matrix convergence, the minimizer of the GMM criterion converges almost surely to β_0 .
- **Main asymptotic-normality result.** If the sample moments satisfy a central limit theorem and the moment function is differentiable around β_0 , then

$$\sqrt{N}(\hat{\beta}_N - \beta_0) \Rightarrow N(0, V),$$

with covariance depending on the long-run variance of the moment process and on the selected linear combinations of moments.

- **Key innovation.** The paper allows the disturbances behind the moment restrictions to be serially correlated and conditionally heteroskedastic, provided the observable process is stationary and ergodic and a suitable long-run variance exists.
- **Optimal weighting.** Among estimators using a fixed finite set of moment restrictions, the optimal weighting matrix uses the inverse of the long-run covariance matrix S_w of the moments. This produces the familiar efficient GMM covariance

$$(d_0' S_w^{-1} d_0)^{-1}.$$

- **Overidentification testing.** If the number of moments r exceeds the number of parameters q , the minimized optimally weighted GMM criterion gives a chi-square test with $r - q$ degrees of freedom.
- **Why the paper matters.** It unifies nonlinear IV, two- and three-stage least squares, least squares, quasi-maximum likelihood first-order conditions, and rational-expectations estimation under one moment-restriction framework.
- **Modern takeaway.** Almost every empirical asset-pricing, macro, labor, and industrial-organization paper using moment restrictions, instruments, HAC covariance matrices, or Hansen's J test is using the framework established here.

2 Paper structure and section-by-section roadmap

2.1 Introduction

The introduction begins from the observation that many economic models imply *orthogonality conditions*. These conditions typically state that model residuals are uncorrelated with instruments, forecasts are uncorrelated with information available at the forecasting date, or Euler-equation errors are orthogonal to variables known to agents. Instead of requiring a complete likelihood, Hansen starts from the restrictions that economic theory is often best able to deliver.

Interpretation: The paper’s methodological shift is important. Rather than forcing the researcher to specify the full distribution of the data, it shows that large-sample estimation and inference can be based on valid population moments alone.

The introduction also explains why standard asymptotic theory is not enough. Many macro-finance applications involve overlapping forecast errors, rational expectations, and conditional heteroskedasticity. In such settings, disturbances may be serially correlated or have time-varying conditional variances. The paper therefore builds an asymptotic theory for stationary and ergodic time series with potentially complicated covariance structure.

Interpretation: This is the bridge between econometric theory and applications. The GMM estimator is valuable not only because it is general, but also because it supports inference when the simple iid textbook covariance formulas are wrong.

2.2 Section 2: Consistency of the GMM estimator

Section 2 defines the first version of the GMM estimator as the minimizer of a quadratic distance from sample moments to zero. It then establishes strong consistency under two alternative sets of conditions.

- The first consistency theorem assumes a compact parameter space, first-moment continuity of the moment function, and global identification.
- The second consistency theorem relaxes compactness for a special class of moment functions that are linear in transformed parameters.
- The third theorem covers recursive estimation, where a subset of parameters is estimated first and the remaining parameters are estimated in a second step.

Interpretation: Section 2 is about existence and convergence. It answers: when does the sample criterion concentrate around a population criterion with a unique minimizer at the true parameter?

2.3 Section 3: Asymptotic distribution of the GMM estimator

Section 3 changes perspective. Instead of defining GMM only as a minimization problem, it defines a generic GMM estimator through first-order conditions: it sets q linear combinations of the r sample moment conditions approximately equal to zero. This allows Hansen to include estimators whose first-order conditions have a moment-restriction interpretation, including least squares and quasi-maximum likelihood.

Interpretation: This section is the central asymptotic-normality section. It formalizes how the choice of weighting matrix affects the asymptotic covariance matrix.

The section introduces the long-run covariance matrix

$$S_w = \sum_{j=-\infty}^{\infty} E[w_t w'_{t-j}], \quad w_t = f(x_t, \beta_0),$$

which is the spectral density matrix of the moment process at frequency zero.

Interpretation: In iid models, only $E[w_t w'_t]$ matters. In time series GMM, all serial covariances matter. This is the origin of HAC-style inference in the GMM framework.

2.4 Section 3 special cases: computing the long-run variance

Hansen then considers five cases for moment functions of the instrumental-variables form

$$f(x_t, \beta_0) = u_t \otimes z_t,$$

where u_t is a disturbance vector and z_t is an instrument vector.

- Case (i) is homoskedastic martingale-difference behavior: S_w factors into disturbance and instrument second moments.
- Case (ii) allows conditional heteroskedasticity: S_w becomes $E[u_t u_t' \otimes z_t z_t']$.
- Case (iii) allows finite-order serial dependence in the moment process, so S_w is a finite autocovariance sum.
- Case (iv) adds constant conditional autocovariances, allowing S_w to be written using disturbance and instrument autocovariances.
- Case (v) considers linearly regular processes and spectral representations.

Interpretation: These cases show how one moves from simple cross-sectional robust variance formulas to time-series HAC formulas, depending on the dependence structure in the moments.

2.5 Section 3 optimality

The optimality subsection asks: among all estimators that use the same finite set of moment conditions, which weighting matrix gives the smallest asymptotic covariance matrix? The answer is the matrix proportional to

$$d_0' S_w^{-1}, \quad d_0 = E \left[\frac{\partial f(x_t, \beta_0)}{\partial \beta'} \right].$$

Interpretation: Efficient GMM is not efficiency among all possible estimators. It is efficiency among estimators that use the specified finite collection of moment restrictions.

2.6 Section 4: Testing overidentifying restrictions

Section 4 derives the test of overidentifying restrictions. If $r > q$, estimation uses only q effective linear combinations of the r moments. The remaining $r - q$ moment restrictions can be tested. With optimal weighting, the statistic

$$N g_N(\hat{\beta}_N)' \hat{S}^{-1} g_N(\hat{\beta}_N)$$

converges to a chi-square distribution with $r - q$ degrees of freedom.

Interpretation: This is the origin of the Hansen J test. It asks whether the moments that were not mechanically forced to zero are still close to zero in the sample.

2.7 Section 5: Conclusion

The conclusion emphasizes that moment-based estimation is useful because it compares and unifies many estimators that exploit the same orthogonality conditions. It also emphasizes the role of stationarity and ergodicity: these assumptions are powerful enough to allow serial correlation and heteroskedasticity but restrictive enough to deliver clean large-sample theory.

Interpretation: The conclusion is modest but foundational. Hansen notes that relaxing stationarity would be valuable but likely requires more complicated regularity conditions.

3 Data, notation, and measurement in the theory

This is not an empirical paper with a dataset in the usual sense. Its “data” are a generic stochastic process of observables. The notation is therefore the model’s measurement system.

3.1 Observable process

The econometrician observes a finite sample

$$x_1, x_2, \dots, x_N,$$

where each x_t is a p -dimensional vector. The underlying process is assumed to be stationary and ergodic.

- **Stationarity** means the joint distribution of the process is time invariant.
- **Ergodicity** means sample averages converge to population expectations.
- These assumptions allow the sample moments $g_N(\beta)$ to converge to their population counterparts.

Interpretation: Stationarity and ergodicity replace iid sampling. They are broad enough for many time-series models but strong enough to justify laws of large numbers.

3.2 Parameter space

The parameter vector is

$$\beta_0 \in S \subseteq \mathbb{R}^q.$$

Section 2 allows S to be a general separable metric space, including compactifications of $\mathbb{R}^q$. Section 3 specializes to an open subset of $\mathbb{R}^q$ containing β_0 because differentiability is needed for asymptotic normality.

Interpretation: Compactness helps consistency because it prevents minimizers from escaping to infinity. When compactness is dropped, Hansen requires additional structure on f .

3.3 Moment function

The moment function is

$$f : \mathbb{R}^p \times S \rightarrow \mathbb{R}^r,$$

with $r \geq q$. The population restriction is

$$E[f(x_t, \beta_0)] = 0.$$

The sample analogue is

$$g_N(\beta) = \frac{1}{N} \sum_{t=1}^N f(x_t, \beta).$$

Interpretation: GMM is useful because r can exceed q . More moments than parameters gives overidentification and hence testable restrictions.

3.4 Instrumental-variables example

A common moment restriction comes from

$$u_t = F(x_t, \beta_0), \quad z_t = G(x_t, \beta_0),$$

with

$$E[u_t \otimes z_t] = 0.$$

Then

$$f(x_t, \beta_0) = F(x_t, \beta_0) \otimes G(x_t, \beta_0).$$

Interpretation: This nests linear and nonlinear IV. The residual u_t need not be iid. What matters is the orthogonality between residuals and instruments.

3.5 Weighting matrix and distance

Let a_N be an $s \times r$ random weighting matrix converging to a_0 . The criterion is

$$Q_N(\beta) = \|a_N g_N(\beta)\|^2.$$

Equivalently, if $W_N = a_N' a_N$,

$$Q_N(\beta) = g_N(\beta)' W_N g_N(\beta).$$

Interpretation: The weighting matrix determines what it means for moments to be close to zero. For consistency, many choices work. For efficiency, the choice is crucial.

3.6 Long-run covariance of the moment process

Define

$$w_t = f(x_t, \beta_0), \quad R_w(j) = E[w_t w_{t-j}'].$$

The long-run covariance is

$$S_w = \sum_{j=-\infty}^{\infty} R_w(j).$$

Interpretation: S_w is the variance of $\sqrt{N} g_N(\beta_0)$ in the presence of serial correlation. If the moments are uncorrelated over time, $S_w = R_w(0)$.

4 Models and theoretical specifications

4.1 Basic population moment model

The theoretical model is simply a set of valid moment restrictions:

$$E[f(x_t, \beta_0)] = 0. \tag{M}$$

Identification requires that these restrictions uniquely identify β_0 , usually in the sense that

$$E[f(x_t, \beta)] = 0 \implies \beta = \beta_0. \tag{1}$$

Interpretation: The paper separates identification from estimation. Identification is a population condition. GMM estimation is the sample analogue.

4.2 Sample moment criterion

The sample criterion is

$$Q_N(\beta) = \|a_N g_N(\beta)\|^2, \quad g_N(\beta) = \frac{1}{N} \sum_{t=1}^N f(x_t, \beta). \quad (\text{C})$$

The estimator solves

$$\hat{\beta}_N \in \arg \min_{\beta \in S} Q_N(\beta). \quad (\text{GMM})$$

Interpretation: The estimator chooses the parameter vector that minimizes the weighted distance between sample moments and zero.

4.3 Population criterion

The population analogue is

$$Q_0(\beta) = \|a_0 E[f(x_t, \beta)]\|^2. \quad (2)$$

If $Q_0(\beta)$ has a unique minimum at β_0 , and if Q_N converges uniformly to Q_0 , then the minimizer of Q_N converges to β_0 .

Interpretation: This is the standard extremum-estimator logic. The paper's contribution is to provide sufficient conditions for the moment-based criterion under stationarity and ergodicity.

4.4 First-order-condition representation

For asymptotic normality, Hansen uses a broader representation. A generic GMM estimator satisfies

$$\sqrt{N} a_N^* g_N(\hat{\beta}_N) \xrightarrow{p} 0, \quad (\text{FOC})$$

where a_N^* is a $q \times r$ matrix of weights converging to a^* .

Interpretation: A minimizer of the quadratic criterion satisfies this representation through its first-order conditions, but other estimators can also be included if their first-order conditions can be written in moment form.

4.5 Linearization around the true parameter

Let

$$d_0 = E \left[\frac{\partial f(x_t, \beta_0)}{\partial \beta'} \right]. \quad (3)$$

A first-order Taylor expansion gives

$$g_N(\hat{\beta}_N) \approx g_N(\beta_0) + d_0(\hat{\beta}_N - \beta_0). \quad (4)$$

Combining this with the first-order-condition representation implies

$$\sqrt{N}(\hat{\beta}_N - \beta_0) \approx -(a^* d_0)^{-1} a^* \sqrt{N} g_N(\beta_0). \quad (5)$$

Interpretation: The estimator inherits its limiting distribution from the sample moments. The derivative matrix d_0 maps changes in parameters into changes in moment conditions.

4.6 Asymptotic normality

If

$$\sqrt{N}g_N(\beta_0) \Rightarrow N(0, S_w), \quad (6)$$

then

$$\sqrt{N}(\hat{\beta}_N - \beta_0) \Rightarrow N\left(0, (a^*d_0)^{-1}a^*S_w a^{*'} [(a^*d_0)^{-1}]'\right). \quad (\text{AN})$$

Interpretation: The asymptotic covariance has three ingredients: the selected moment combinations a^* , the sensitivity of moments to parameters d_0 , and the long-run covariance of the moments S_w .

4.7 Optimal GMM weighting

The optimal limiting first-order-condition weight is proportional to

$$a^* = e d_0' S_w^{-1}, \quad (\text{OPT})$$

where e is any nonsingular $q \times q$ matrix. The resulting covariance is

$$V_{\text{opt}} = (d_0' S_w^{-1} d_0)^{-1}. \quad (\text{EV})$$

Interpretation: The nonsingular matrix e changes the normalization of the equations but not the estimator's asymptotic variance. Efficient GMM uses inverse long-run covariance weighting.

4.8 Two-step feasible GMM

A practical version proceeds as follows:

1. Estimate β_0 using any consistent weighting matrix.
2. Use residual moments $f(x_t, \hat{\beta})$ to estimate S_w .
3. Re-estimate using $\widehat{W} = \widehat{S}^{-1}$.

Interpretation: This is the feasible GMM algorithm used in empirical work. The first step need not be efficient; it only needs to be consistent enough to estimate the optimal weight.

4.9 Overidentification test

When $r > q$, define the optimally weighted minimized criterion statistic

$$J_N = N g_N(\hat{\beta}_N)' \widehat{S}^{-1} g_N(\hat{\beta}_N). \quad (\text{J})$$

Under the model and optimal weighting,

$$J_N \Rightarrow \chi_{r-q}^2. \quad (7)$$

Interpretation: The J statistic tests whether all moment restrictions are jointly compatible with the data. It is not a test of parameter estimation alone; it is a test of the model's overidentifying restrictions.

5 Identification, assumptions, and credibility of the theory

5.1 Identification

Identification requires that the population moments distinguish β_0 from other parameter values. In Hansen's compact-parameter-space theorem, the key condition is that

$$h_0(\beta) = a_0 E[f(x_t, \beta)]$$

has a unique zero at β_0 .

Interpretation: If the population moments are zero at multiple parameter values, no sample estimator based only on those moments can consistently choose among them.

5.2 Why stationarity and ergodicity matter

The sample criterion depends on sample averages. Stationarity and ergodicity ensure that these sample averages converge to their expectations:

$$\frac{1}{N} \sum_{t=1}^N f(x_t, \beta) \rightarrow E[f(x_t, \beta)].$$

Interpretation: The paper's law-of-large-numbers foundation is time-series compatible. This is why it is especially useful for macroeconomics and finance.

5.3 Moment continuity

For consistency, Hansen requires continuity conditions that prevent small changes in β from producing large discontinuous changes in $f(x_t, \beta)$. The paper formalizes this with first-moment continuity.

Interpretation: Moment continuity is the technical condition that supports uniform convergence of the criterion. Without it, minimizing the sample criterion may not mimic minimizing the population criterion.

5.4 Differentiability and rank

For asymptotic normality, the derivative matrix

$$d_0 = E \left[\frac{\partial f(x_t, \beta_0)}{\partial \beta'} \right]$$

must exist and have full rank.

Interpretation: Full rank means the moments have enough local information to identify all coordinates of β_0 . If d_0 is rank deficient, local parameter changes may leave moments unchanged.

5.5 Long-run variance and the central limit theorem

The asymptotic distribution relies on a central limit theorem for the stationary ergodic moment process $w_t = f(x_t, \beta_0)$. Hansen's Assumption 3.5 is designed to ensure that the long-run covariance

$$S_w = \sum_{j=-\infty}^{\infty} R_w(j)$$

is finite and that $\sqrt{N}g_N(\beta_0)$ is asymptotically normal.

Interpretation: This condition is the heart of robust time-series inference. It tells the researcher to estimate the variance of sample moments, not the variance of residuals alone.

5.6 What the paper does not do

- It does not choose the economic moment restrictions; the researcher must specify them.
- It does not solve efficiency among all possible instruments or all possible moments; optimality is conditional on a finite set of given moments.
- It does not provide a fully developed nonstationary GMM theory.
- It does not give finite-sample guarantees; the results are large-sample.
- It does not eliminate the need for economically credible instruments. Invalid moments lead to invalid estimates and invalid J tests.

Interpretation: Hansen's contribution is foundational but not automatic. Empirical success still depends on valid orthogonality restrictions and accurate covariance estimation.

6 Original-paper excerpts: equations, assumptions, theorems, and tests

The paper has no empirical tables or plotted figures. To preserve the original-paper material in the same note style, the key assumptions, theorem statements, and equation blocks are directly cropped from the paper and grouped compactly.

6.1 Title/abstract + orthogonality setup

Read:

- The abstract states the core aim: study estimators that make sample analogues of population orthogonality conditions close to zero.
- The first formal setup shows how residual-instrument orthogonality becomes a moment function $f(x_t, \beta_0)$.
- The paper begins from moment restrictions rather than likelihood assumptions.

Economic message: GMM turns economic theory into estimation by converting model-implied orthogonality restrictions into sample moment equations.

Econometrica, Vol. 50, No. 4 (July, 1982)

LARGE SAMPLE PROPERTIES OF GENERALIZED METHOD OF
MOMENTS ESTIMATORS¹

BY LARS PETER HANSEN

This paper studies estimators that make sample analogues of population orthogonality conditions close to zero. Strong consistency and asymptotic normality of such estimators is established under the assumption that the observable variables are stationary and ergodic. Since many linear and nonlinear econometric estimators reside within the class of estimators studied in this paper, a convenient summary of the large sample properties of these estimators including some whose large sample properties have not heretofore been

(a) Title, abstract, and the paper's central claim.

A common way to obtain orthogonality conditions is to exploit the assumption that disturbances in an econometric model are orthogonal to functions of a set of variables that the econometrician observes. For example, suppose that the econometric model is given by

$$(1) \quad \begin{aligned} u_n &= F(x_n, \beta_0), \\ z_n &= G(x_n, \beta_0), \end{aligned}$$

where

$$(2) \quad E[u_n \otimes z_n] = 0.$$

The vector functions F and G are specified *a priori*, u_n is an unobservable vector of disturbance terms, z_n is a vector of instrumental variables, and " $\otimes$ " denotes the Kronecker product. The dependence of G on its second argument is often-times trivial. When (2) is satisfied, we can let the function f be given by

$$(3) \quad f(x_n, \beta_0) = F(x_n, \beta_0) \otimes G(x_n, \beta_0),$$

and it follows that

$$E[f(x_n, \beta_0)] = 0.$$

(b) Moment restrictions from residual-instrument orthogonality.

6.2 Criterion-function definition

Read:

- The sample moment function is $g_N(\beta) = N^{-1} \sum_t f(x_t, \beta)$.
- The GMM criterion is $\|a_N g_N(\beta)\|^2$.
- The estimator is any measurable element of the set of minimizers.

Economic message: Estimation is a minimum-distance problem: choose the parameter that makes the model-implied sample moments as close to zero as possible.

estimator of the unknown parameter vector β_0 .

For our discussion of consistency, we introduce a sequence of random weighting matrices $\{a_N : N \geq 1\}$ that are dimensioned s by r where $q \leq s \leq r$. The matrices are random in order to allow for their possible dependence on sample information.

ASSUMPTION 2.5: The sequence of random matrices $\{a_N : N \geq 1\}$ converges almost surely to a constant matrix a_0 .⁶

These weighting matrices are used in conjunction with a method of moments estimator of $E[f(x_n, \beta)]$ to obtain a sample objective function whose minimizer is our estimator of β_0 . Let

$$\begin{aligned} f_n(\omega, \beta) &= f[x_n(\omega), \beta], \\ g_N(\omega, \beta) &= \frac{1}{N} \sum_{n=1}^N f_n(\omega, \beta), \\ h_N(\omega, \beta) &= a_N(\omega) g_N(\omega, \beta), \\ B_N(\omega) &= \left\{ \beta \in S : |h_N(\omega, \beta)|^2 = \inf_{\beta \in S} |h_N(\omega, \beta)|^2 \right\}. \end{aligned}$$

6.3 Consistency: compact case and noncompact extension

Read:

- The compact consistency theorem requires stationarity/ergodicity, moment continuity, compactness, and unique identification.
- The noncompact extension works for a special structure $f(x, \beta) = c_0(x) + c_1(x)X(\beta)$.
- The distinction is a tradeoff: weaker parameter-space compactness requires stronger structure on the moment function.

Economic message: Strong consistency follows from uniform convergence of the sample criterion to a population criterion with a unique zero.

We now consider our first consistency theorem for the GMM estimator.

THEOREM 2.1: Suppose Assumptions 2.1–2.5 are satisfied. If (i) f_1 is first moment continuous for all $\beta \in S$; (ii) S is compact; (iii) $h_0(\beta)$ has a unique zero at β_0 ; then a GMM estimator $\{b_N : N \geq 1\}$ exists and converges almost surely to β_0 .

Condition (iii) of this theorem is the parameter identification requirement that the population orthogonality conditions used in estimation be satisfied only at the true parameter vector. When a_0 is an r by r nonsingular matrix, h_0 will have a unique zero at β_0 if, and only if, $Ef(x_1, \cdot)$ has a unique zero at β_0 . When a_0 has fewer rows than columns ($s < r$), condition (iii) imposes the more stringent requirement that s linear combinations of the population orthogonality conditions are satisfied only at the true parameter vector. For this reason, it may be judicious to choose a_0 to be an r by r nonsingular matrix.¹¹

The compactness condition (ii) of Theorem 2.1 can be weakened if a special structure is imposed on the function f . Consider the following assumption.

ASSUMPTION 2.6: $f(x_1, \beta) = c_0(x_1) + c_1(x_1)\lambda(\beta)$ where $c_0(x_1)$ is an r dimensional column vector, $c_1(x_1)$ is an r by m matrix, and $\lambda(\beta)$ is an m dimensional vector.

(a) Theorem 2.1: compact-space consistency.

This assumption accommodates models that are linear in the variables but not necessarily in the parameters. Our next theorem establishes consistency for models with orthogonality conditions that satisfy Assumption 2.6.

THEOREM 2.2: Suppose Assumptions 2.1–2.6 are satisfied. If (i) (S, σ) is locally compact; (ii) λ is continuous on S , and for any $\phi > 0$ there exists a $\rho > 0$ such that $|\lambda(\alpha) - \lambda(\beta)| < \rho$, $\sigma(\alpha, \beta) < \phi$; (iii) for any $\rho > 0$,

$$\inf \left\{ \frac{1}{1 + |\lambda(\beta)|} |h(\beta)| : \beta \in S, |\lambda(\beta) - \lambda(\beta_0)| > \rho \right\} > 0;$$

then the GMM estimator $\{b_N : N \geq 1\}$ exists and converges almost surely to β_0 .

(b) Theorem 2.2 and related noncompact logic.

6.4 Recursive GMM estimation

Read:

- The recursive result allows one subset of parameters to be estimated first.
- The remaining parameters are then estimated conditional on the first-step consistent estimate.
- This is useful computationally when the full criterion is hard to minimize jointly.

Economic message: Consistency can survive sequential estimation when the first step is consistent and the parameter space permits convergence back to the true full parameter vector.

$\beta_{1,0}$, we can construct an estimator of $\beta_{2,0}$ by minimizing

$$|h_N[\omega, b_{1,N}(\omega), \beta_2]|^2$$

with respect to β_2 such that $(\beta'_1, \beta'_2)' \in S$, where $\beta_1 = b_{1,N}(\omega)$. Theorem 2.3 establishes the consistency of this recursive estimator.

THEOREM 2.3: *Suppose Assumptions 2.1–2.5 are satisfied. If (i) the conditions of Theorem 3.1 are satisfied; (ii) $\{b_{1,N} : N \geq 1\}$ converges almost surely to $\beta_{1,0}$; (iii) for any sequence $\{\gamma_j : j \geq 1\}$ in S such that $\{\gamma_{1,j} : j \geq 1\}$ converges to $\beta_{1,0}$, there exists a sequence $\{\beta_{2,j} : j \geq 1\}$ such that $\{(\gamma_{1,j}, \beta_{2,j}) : j \geq 1\}$ is a sequence in S that converges to β_0 ; then a GMM estimator $\{b_N : N \geq 1\}$ exists and converges almost surely to β_0 .¹⁴*

6.5 Asymptotic assumptions and generic GMM definition

Read:

- Section 3 assumes an open parameter space, differentiability, full-rank derivative matrix, and a CLT-compatible moment process.
- Definition 3.1 defines GMM through asymptotically satisfied linear combinations of sample moments.
- Lemma 3.2 connects the minimization-based estimator to the generic first-order-condition form.

Economic message: The first-order-condition view is what makes GMM a unifying framework for many estimators beyond explicit minimum-distance estimators.

We require that a GMM estimator $\{b_N^*: N \geq 1\}$ asymptotically satisfy the set equations

$$a_0^* E f(x_n, \beta) = 0$$

in the sense of Definition 3.1.

DEFINITION 3.1: The GMM estimator $\{b_N^*: N \geq 1\}$ is a sequence of random vectors that converges in probability to β_0 for which $\{\sqrt{N}a_N^*g_N(b_N^*): N \geq 1\}$ converges in probability to zero.

Before showing that this GMM estimator is asymptotically normal and displaying the dependence of its asymptotic covariance matrix on the limiting weighting matrix a_0^* , we discuss the link between this estimator and the GMM estimator of Definition 2.1. Note that

$$|h_N(\beta)|^2 = |a_N g_N(\beta)|^2 = g_N(\beta)' a_N' a_N g_N(\beta).$$

Assuming that the first-order conditions for the problem of minimizing $|h_N|^2$ are satisfied by b_N , then

$$(6) \quad \frac{\partial g_N}{\partial \beta}(b_N)' a_N' a_N g_N(b_N) = 0.$$

Let a_N^* be the q by r matrix

$$(7) \quad a_N^* = \frac{\partial g_N}{\partial \beta}(b_N)' a_N' a_N.$$

Substituting (6) into (7), we obtain $a_N^* g_N(b_N) = 0$ which trivially satisfies one of the key requirements of Definition 2.1. Once we establish the strong consistency of the estimator of Definition 2.1, require that the first-order conditions (6) be satisfied, and demonstrate that the sequence $\{a_N^*: N \geq 1\}$ converges in probability to a constant matrix, then we obtain a GMM estimator of Definition 3.1. Lemma 3.2 supplies sufficient conditions for $\{a_N^*: N \geq 1\}$ as defined by (7) to converge in probability to a constant matrix.

LEMMA 3.2: Suppose Assumptions 3.1–3.4 are satisfied. If (i) $\{b_N: N \geq 1\}$ converges in probability to β_0 ; (ii) $\{a_N: N \geq 1\}$ converges in probability to a_0 ; then $\{(\partial g_N / \partial \beta)(b_N): N \geq 1\}$ converges in probability to d_0 and $\{a_N^*: N \geq 1\}$ given by (7) converges in probability to $a_0^* = d_0' a_0$.

We begin our asymptotic distribution discussion by describing the underlying assumptions which we make. We extend the index set of the stochastic process containing the observable variables from the nonnegative integers to include all of the integers. For studying probabilistic properties, Doob [7, p. 456] argues that this extension is innocuous.

ASSUMPTION 3.1: $\{x_n: -\infty < n < +\infty\}$ is stationary and ergodic.

We modify the specification and role of the set S in the analysis.

ASSUMPTION 3.2: S is an open subset of R^q that contains β_0 .

We use the metric implied by the absolute value norm to define our notion of convergence on S . We place additional requirements on the function f and the process $\{x_n: -\infty < n < +\infty\}$.

ASSUMPTION 3.3: $f(\cdot, \beta)$ and $\partial f / \partial \beta(\cdot, \beta)$ are Borel measurable for each $\beta \in S$ and $\partial f / \partial \beta(x, \cdot)$ is continuous on S for each $x \in R^p$.

ASSUMPTION 3.4: $\partial f_1 / \partial \beta$ is first moment continuous at β_0 , and $E[\partial f / \partial \beta(x_1, \beta_0)]$ exists, is finite, and has full rank.

(a) Assumptions for asymptotic normality.

(b) Definition 3.1 and link to first-order conditions.

6.6 Theorem 3.1: asymptotic normality

Read:

- Theorem 3.1 gives the limiting normal distribution of $\sqrt{N}(\hat{\beta}_N - \beta_0)$.
- The covariance matrix depends on a^* , d_0 , and S_w .
- This formula covers serial correlation and conditional heteroskedasticity through S_w .

Economic message: Standard errors must be built from the long-run variance of the moment conditions. The estimator's asymptotic distribution is not generally the iid IV formula.

In preparation for our asymptotic distribution theorem, we let

$$R_w(j) = E[w_0 w'_{-j}].$$

Assumptions (3.1) and (3.5) insure that $R_w(j)$ is finite and that the matrix

$$S_w = \sum_{j=-\infty}^{+\infty} R_w(j)$$

is well defined and finite.¹⁹ Theorem 3.1 displays the asymptotic distribution of the GMM estimator.

THEOREM 3.1: *Suppose Assumptions 3.1–3.6 are satisfied. Then $\{\sqrt{N}(b_N^* - \beta_0) : N \geq 1\}$ converges in distribution to a normally distributed random vector with mean zero and covariance matrix $(a_0^* d_0)^{-1} a_0^* S_w a_0^* (a_0^* d_0)^{-1}$.²⁰*

Since S_w plays a prominent role in the expression for the asymptotic covariance matrix, we shall examine Assumption (3.5) in conjunction with the computation of S_w . We focus on situations in which

$$(10) \quad f(x_n, \beta_0) = u_n \otimes z_n$$

where we view z_n as a vector of the instrumental variables and u_n is a vector of the disturbance terms from the econometric model. Let

$$R_u(j) = E[u_n u'_{n-j}]$$

and

$$R_z(j) = E[z_n z'_{n-j}]$$

and assume that $R_u(0)$ and $R_z(0)$ exist and are finite. It is instructive for us to examine five special cases.

6.7 Special cases for the long-run covariance matrix

Read:

- Cases (i) and (ii) handle martingale-difference moments, with and without conditional homoskedasticity.
- Cases (iii) and (iv) handle finite serial correlation, especially relevant for overlapping forecasts and rational-expectations models.
- Case (v) uses spectral-density representations for more general time-series dependence.

Economic message: These cases are a menu for empirical covariance estimation. The right standard errors depend on which dependence structure is credible.

CASE (i):

$$E[u_n | z_n, u_{n-1}, z_{n-1}, u_{n-2}, \dots] = 0,$$

$$E[u_n u'_n | z_n, u_{n-1}, z_{n-1}, u_{n-2}, \dots] = R_u(0).$$

This case includes as a special subcase models in which $\{u_n : -\infty < n < +\infty\}$ is a sequence of independent, random vectors and u_n is independent of $\{z_j : -\infty < j \leq n\}$. It is straightforward to verify that

$$E[w_0 | w_{-j}, w_{-j-1}, \dots] = 0 \quad \text{for } j \geq 1,$$

$$v_0 = w_0,$$

and

$$v_j = 0 \quad \text{for } j \geq 1.$$

This shows that Assumption 3.5 is satisfied. Also, it can be demonstrated that

$$R_w(j) = 0 \quad \text{for } j \neq 0$$

and

$$S_w = R_w(0) = R_u(0) \otimes R_z(0).$$

Thus, S_w can be computed from the second moments of z_n and u_n .

CASE (ii):

$$E[u_n | z_n, u_{n-1}, z_{n-1}, u_{n-2}, \dots] = 0.$$

This case differs from Case (i) in that we no longer assume that the conditional covariance matrix for u_n is independent of the conditioning set. This allows for a particular form of heteroskedasticity. The stationarity assumption, however, restricts us to circumstances in which the unconditional variances of $\{u_n : -\infty < n < +\infty\}$ are constant. As in Case (i), it can be verified that Assumption 3.5 of Theorem 3.1 is met and that

$$R_w(j) = 0 \quad \text{for } j \neq 0.$$

In contrast with Case (i), we can no longer compute $R_w(0)$, and consequently S_w , from the second moments of u_n and z_n . More specifically, we have

$$S_w = R_w(0) = E[u_n u'_n \otimes z_n z'_n].$$

This form of S_w arises in the multiperiod probit estimators proposed by Avery, Hansen, and Hotz [3].

CASE (iii):

$$E[u_n | z_n, u_{n-k}, z_{n-1}, u_{n-k-1}, \dots] = 0.$$

(a) Cases (i)–(iv): martingale and finite-dependence structures.

CASE (v): Suppose that the $\{(u'_n, z'_n) : -\infty < n < +\infty\}$ process is linearly regular and has fourth order cumulants that are zero. For simplicity we assume that

$$E[z_n] = 0,$$

$$E[u_n] = 0.$$

Let

$$E[u_n z'_{n-j}] = \begin{bmatrix} R_{uz}^1(j) \\ R_{uz}^2(j) \\ \vdots \\ R_{uz}^L(j) \end{bmatrix}$$

where L is the number of elements in the disturbance vector and where $R_{uz}^k(j)$ is an M dimensional row vector with the same number of elements as are in the instrument vector. Define

$$\bar{R}_w(j) = \begin{bmatrix} R_{uz}^1(-j)' R_{uz}^1(j) & R_{uz}^2(-j)' R_{uz}^1(j) & \cdots & R_{uz}^L(-j)' R_{uz}^1(j) \\ R_{uz}^1(-j)' R_{uz}^2(j) & R_{uz}^2(-j)' R_{uz}^2(j) & \cdots & R_{uz}^L(-j)' R_{uz}^2(j) \\ \vdots & \vdots & \ddots & \vdots \\ R_{uz}^1(-j)' R_{uz}^L(j) & R_{uz}^2(-j)' R_{uz}^L(j) & \cdots & R_{uz}^L(-j)' R_{uz}^L(j) \end{bmatrix}.$$

Then it can be shown that

$$(8) \quad R_w(j) = \bar{R}_w(j) + R_u(j) \otimes R_z(j),$$

$$S_w = \sum_{j=-\infty}^{+\infty} [\bar{R}_w(j) + R_u(j) \otimes R_z(j)].$$

(b) Case (v): spectral representation.

6.8 Consistent estimation of S_w and optimal weighting

Read:

- The paper explains how to estimate autocovariances of the residual moment process.
- When only finitely many autocovariances matter, S_w can be estimated by a finite sum.
- More generally, S_w is the zero-frequency spectral density of the moment process.
- Theorem 3.2 gives the efficient finite-moment weighting matrix.

Economic message: Feasible efficient GMM is a two-step logic: get a consistent estimate first, estimate the long-run variance of the moments, then re-weight moments by the inverse long-run variance.

LEMMA 3.3: Suppose Assumptions 3.1–3.5 are satisfied. If f_1 is second moment continuous at β_0 , then $\{R_w^N(j): N \geq 1\}$ converges to $R_w(j)$ in probability.

In situations where S_w depends on a finite number of autocovariances, i.e.,

$$S_w = \sum_{j=-k+1}^{k-1} R_w(j),$$

we can use Lemma 3.2 to argue that:

$$S_w^N = \sum_{j=-k+1}^{k-1} R_w^N(j)$$

is a consistent estimator of S_w . Furthermore, if the conditional covariance assumptions of Case (i) or (iv) are met, the special structure of S_w can be exploited even further. Lemma 3.3 can be used to establish consistency of the sample autocovariances of the estimated disturbances and the instruments. In the general case, S_w cannot necessarily be computed from a finite number of autocovariances which complicates its consistent estimation. However, S_w is the spectral density matrix of $\{w_n: -\infty < n < +\infty\}$ at frequency zero, and a consistent estimator of S_w can be obtained by using procedures appropriate for estimating spectral density matrices.²¹

(a) Estimating the long-run covariance of moments.

THEOREM 3.2: Suppose $\{b_N^*: N \geq 1\} \in A$ and that the limiting weighting matrix associated with $\{b_N^*: N \geq 1\}$ satisfies

$$(10) \quad (a_0^* d_0)^{-1} a_0^* S_w a_0^* (a_0^* d_0)^{-1'} = (d_0' S_w^{-1} d_0)^{-1}.$$

Then $\{b_N^*: N \geq 1\}$ is optimal with asymptotic covariance matrix $(d_0' S_w^{-1} d_0)^{-1}$. Furthermore, all optimal estimators in A will have a limiting weighting matrix that satisfies (10), and

$$(11) \quad a_0^* = e d_0' S_w^{-1}$$

for some q by q nonsingular matrix e .

In order to determine an optimal choice of a_0^* of the form specified in (11), it is necessary to have a consistent estimator of d_0 and S_w . This can be accomplished by initially employing a not necessarily optimal GMM estimator and using one of the estimation strategies mentioned earlier for S_w and d_0 .

In considering the GMM estimators of Section 2, we indicated that from the standpoint of consistency it may be desirable to employ a square nonsingular matrix a_0 as the limiting weighting matrix. If we choose a_0 such that $a_0' a_0 = S_w^{-1}$ and a_N is some consistent estimator of a_0 , then Lemma 3.2 informs us that the corresponding a_0^* is equal to $d_0 S_w^{-1}$. Thus the resulting estimator is optimal. Under the assumptions defining Case (i) above, the choice of $a_0' a_0 = S_w^{-1}$ yields the nonlinear instrumental variables estimators discussed by Amemiya [1], Jorgenson and Laffont [24], and Gallant [11].

(b) Theorem 3.2: optimal finite-moment weighting.

6.9 Overidentifying restrictions and Hansen's J test

Read:

- If $r > q$, not all moments are used mechanically to estimate parameters.
- The leftover restrictions can be tested using the optimally weighted minimized criterion.
- Under the null, the statistic converges to χ_{r-q}^2 .

Economic message: The overidentification test checks whether the model's extra moment restrictions are compatible with the data. Rejection means at least some of the assumed orthogonality restrictions fail.

4. TESTING OVER-IDENTIFYING RESTRICTIONS

When the number of orthogonality conditions, r , exceeds the number of parameters to be estimated, q , tests of the restrictions implied by the econometric model are available. As was noted in Section 3, estimation of the model parameters sets q linear combinations of the r sample orthogonality conditions equal to zero, at least asymptotically. Thus, when the model is true, there are $r - q$ linearly independent combinations of the orthogonality conditions that ought to be close to zero but are not actually set to zero. This provides us with a scheme for testing the over-identifying restrictions of the model which is elaborated upon below.

Using the results of Section 3, we can obtain the asymptotic distribution of $\{\sqrt{N}g_N(b_N^*): N \geq 1\}$. Recall that $g_N(b_N^*)$ is an expression for the sample orthogonality conditions evaluated at the parameter estimator b_N^* . Lemma 4.1 provides the desired asymptotic result.

LEMMA 4.1: Suppose Assumptions 3.1–3.6 are satisfied. Then $\{\sqrt{N}g_N(b_N^*): N \geq 1\}$ converges in distribution to a normal random vector with mean zero and covariance matrix $\zeta_0 = [I - d_0(a_0^* d_0)^{-1} a_0^*] S_w [I - d_0(a_0^* d_0)^{-1} a_0^*]'$.

Since we have assumed that $\{\sqrt{N}a_N^* g_N(b_N^*): N \geq 1\}$ converges to zero in probability, it is reasonable to suspect that the asymptotic covariance matrix ζ_0 given in Lemma 4.1 is singular. We can verify this singularity by premultiplying ζ_0 by a_0^* and obtaining a matrix of zeroes. Although ζ_0 is singular, if S_w is nonsingular and r exceeds q , then ζ_0 is not zero. Hence there are linear combinations of the sample orthogonality conditions that have a nondegenerate asymptotic distribution. These linear combinations of sample orthogonality conditions can be used to obtain asymptotically valid test statistics of the model restrictions.

We wish to examine a particularly convenient test statistic of this form. This test can be viewed as an extension of a specification test proposed by Sargan [29, 30] and of the specification test associated with minimum chi-square estimators (see Ferguson [10]). Let

$$\tau_N = g_N(b_N^*) (S_w^N)^{-1} g_N(b_N^*),$$

where $\{S_w^N: N \geq 1\}$ is a consistent estimator of S_w . Its asymptotic distribution is given in Lemma 4.2 assuming $\{b_N^*: N \geq 1\}$ is an optimal estimator as defined in Section 3.

LEMMA 4.2: Suppose Assumptions 3.1–3.6 of Theorem 4.1 are satisfied and that $a_0 = e d_0' S_w^{-1}$ for some q by q nonsingular matrix e . Then $N\tau_N$ converges in

6.10 Conclusion and proof logic

Read:

- The conclusion emphasizes the breadth of the moment-based framework and the role of stationarity and ergodicity.
- The appendix sketches the linearization proof behind asymptotic normality and the algebra behind the optimality and test results.

Economic message: The proof strategy is the standard modern GMM proof: establish consistency, linearize sample moments, apply a CLT to moment averages, and map the result through the derivative and weighting matrices.

5. CONCLUSION

This paper has provided a discussion of the large sample properties of a class of econometric estimators that are defined in terms of orthogonality conditions. Viewing estimation in this way is convenient for comparing estimators that exploit, at least implicitly, the same set of orthogonality conditions and is suggestive of computationally practical estimators in situations in which asymptotically efficient estimation is computationally burdensome. The contribution of this paper is to provide a discussion of consistency and asymptotic normality of estimators under conditions not previously examined by other researchers.

In our discussion we exploited the assumption that the underlying stochastic process of observables is stationary and ergodic. Assumptions of this nature oftentimes play a role in model specification. Lucas [25] and Lucas and Sargent [26] have emphasized that in time series modeling based on dynamic theory, the stochastic properties of the forcing variables play a critical role in model specification. Characterizing the forcing variables as a stationary process is clearly convenient in deriving the dynamic decision rules of economic agents because stationary processes have time invariant probability laws. Furthermore, the stationary assumption and the theorems in this paper accommodate potentially complicated conditional covariance structures for the disturbance terms and the observable variables. On the other hand, there exist many situations in which it would be useful to relax the stationarity assumption. It seems likely, however, that such extensions will either employ more obscure regularity conditions, or will employ regularity conditions that are not uniformly weaker than those used in this paper. Nonetheless, it would be a useful exercise to examine the extent to which results like those obtained here remain intact or could easily be modified when a subset of forcing variables are not stationary (even though they may have a time invariant representation). Along this vein, extensions of the cross-sectional results of White [32, 33, 34] and cross-sectional and time series

PROOF OF THEOREM 3.2: First, we factor $S_w = CC'$ where C is r by r and nonsingular. Second, we let

$$D = (a_0' d_0)^{-1} a_0' C - (d_0' S_w^{-1} d_0)^{-1} d_0' C^{-1}.$$

Third, we note that

$$DC^{-1} d_0 = 0.$$

Fourth, we verify that

$$(a_0' d_0)^{-1} a_0' S_w a_0' (a_0' d_0)^{-1} = DD' + (d_0' S_w^{-1} d_0)^{-1}.$$

Thus, $(d_0' S_w^{-1} d_0)^{-1}$ is a lower bound for the asymptotic covariance matrix of elements in A . This lower bound is attained if and only if $D = 0$.

We can premultiply D by $a_0' d_0$, postmultiply D by C^{-1} , and claim that if $D = 0$, then

$$a_0' - a_0' d_0 (d_0' S_w^{-1} d_0)^{-1} d_0' S_w^{-1} = 0$$

or

$$a_0' = e d_0' S_w^{-1}$$

where

(a) Conclusion excerpt.

(b) Appendix proof sketch.

7 Subsection-by-subsection interpretation

7.1 Why GMM is more general than IV

Instrumental variables is a special case of GMM because IV is based on moment conditions of the form

$$E[z_t u_t(\beta_0)] = 0.$$

GMM keeps this logic but allows $u_t(\beta)$ to be nonlinear, z_t to be any vector of functions of observables, and the covariance matrix of $z_t u_t$ to be serially correlated and heteroskedastic.

Interpretation: IV estimates coefficients by making instrument-residual correlations vanish. GMM estimates parameters by making any valid model-implied moment functions vanish.

7.2 Why the weighting matrix does not determine consistency

For consistency, the weighting matrix mainly needs to converge and not destroy identification. If $a_0 E[f(x_t, \beta)]$ has a unique zero at β_0 , the minimizer can be consistent even if the weight is not efficient.

Interpretation: Bad weighting can reduce efficiency, but it does not necessarily create inconsistency. Invalid moments or failure of identification create inconsistency.

7.3 Why optimal weighting uses inverse long-run variance

Moments with high variance are noisy. Moments highly correlated with each other contain redundant information. Weighting by S_w^{-1} downweights noisy moments and accounts for covariance among moments.

Interpretation: Efficient GMM is generalized least squares applied to the moment equations.

7.4 Why J tests require overidentification

If $r = q$, the model has exactly as many moment restrictions as parameters. The estimator can usually set the sample moments to zero exactly, leaving no unused restrictions to test. If $r > q$, only q dimensions are used to estimate q parameters, leaving $r - q$ dimensions for specification testing.

Interpretation: The J test is not available because GMM is used; it is available because the model has more valid moments than parameters.

7.5 Why conditional heteroskedasticity matters

In a heteroskedastic moment model, even if $E[u_t | z_t] = 0$, the variance of u_t can depend on z_t . Then the correct covariance of the moment $u_t \otimes z_t$ is

$$E[u_t u_t' \otimes z_t z_t'],$$

not $E[u_t u_t'] \otimes E[z_t z_t']$.

Interpretation: Treating conditionally heteroskedastic moments as homoskedastic gives wrong weights and wrong standard errors.

7.6 Why serial correlation matters

If moments are serially correlated, the variance of the sample average includes autocovariances:

$$\text{Var} \left(\sqrt{N} g_N(\beta_0) \right) = R_w(0) + \sum_{j=1}^{\infty} \{R_w(j) + R_w(j)'\}.$$

Interpretation: This is why overlapping-return regressions, rational-expectations forecast errors, and Euler-equation models need HAC-style inference.

7.7 How to read the asymptotic covariance formula

The covariance formula

$$(a^* d_0)^{-1} a^* S_w a^{*'} [(a^* d_0)^{-1}]'$$

can be read from right to left.

- S_w measures the uncertainty in the raw moment average.

- a^* selects q combinations of the r moments.
- d_0 converts parameter movement into moment movement.
- $(a^*d_0)^{-1}$ converts moment uncertainty back into parameter uncertainty.

Interpretation: The estimator is asymptotically normal because the moments are asymptotically normal and the estimator is locally a linear function of them.

7.8 How to implement Hansen's GMM in practice

A practical empirical workflow is:

1. Specify moment restrictions $E[f(x_t, \beta_0)] = 0$ from the model.
2. Choose an initial weighting matrix, often identity or a simple robust matrix.
3. Estimate $\hat{\beta}_1$ by minimizing $g_N(\beta)'W_0g_N(\beta)$.
4. Compute residual moments $\hat{w}_t = f(x_t, \hat{\beta}_1)$.
5. Estimate S_w using a heteroskedastic or HAC estimator depending on the application.
6. Re-estimate with $W = \hat{S}^{-1}$.
7. Report standard errors based on $(\hat{d}'\hat{S}^{-1}\hat{d})^{-1}/N$.
8. If $r > q$, report the J statistic and its χ^2_{r-q} reference distribution.

Interpretation: The modern empirical GMM recipe is essentially Section 3 plus Section 4 of this paper.

8 Short summary

- Hansen (1982) formalizes generalized method of moments estimation for stationary and ergodic processes.
- The estimator is based on population moment restrictions $E[f(x_t, \beta_0)] = 0$.
- The sample criterion minimizes the weighted distance between sample moments and zero.
- Strong consistency follows under moment continuity, identification, and weighting-matrix convergence.
- Compactness of the parameter space is sufficient but not always necessary.
- Recursive GMM estimators can be consistent if the first-step estimator is consistent and the parameter-space conditions work.
- Asymptotic normality follows by linearizing moments around β_0 and applying a CLT to the moment process.
- The asymptotic variance depends on the long-run covariance matrix S_w of the sample moments.
- Serial correlation and conditional heteroskedasticity are handled through S_w .
- Efficient finite-moment GMM uses the inverse long-run covariance matrix as the weighting matrix.
- The efficient GMM covariance is $(d_0'S_w^{-1}d_0)^{-1}$.
- When there are more moment restrictions than parameters, the minimized optimally weighted criterion yields a chi-square overidentification test.
- The framework unifies nonlinear IV, least squares, quasi-MLE first-order conditions, and rational-expectations estimation.
- The main limitation is that the moment restrictions must be valid and the theory is large-sample and stationary/ergodic.

9 Conclusion

9.1 Core conclusion

Hansen's paper establishes the large-sample theory for estimators defined by orthogonality conditions. The paper proves consistency, asymptotic normality, optimal weighting, and overidentification testing for a broad class of estimators under stationary and ergodic sampling.

9.2 Economic intuition

Economic theory often implies that certain variables should be uncorrelated with certain residuals or pricing errors. GMM turns those orthogonality restrictions into a statistical estimator. The researcher does not need to specify a full likelihood, but must correctly specify the moments and estimate their long-run covariance.

9.3 Takeaway

The paper is foundational because it converts the method of moments into a general econometric technology for time series and cross-sectional applications. Its most durable lessons are: write down valid moments, minimize their weighted distance from zero, use inverse long-run covariance weighting for efficiency, and test extra moments with the J statistic.